

Functions:

for every x value, there is only one y value paired

$$f(x) = x^2$$

we get a point as a result $(x, f(x))$

$$x = 2$$

$$f(x) = x^2 = 4$$

$$(2, 4)$$

Function fitting:

we want to get a line which goes through our points

$A(2, 3)$, $B(8, 5)$

General formula:

$$f(x) = a \cdot x + b$$

where a is the slope of the function

and b is where the function intercepts the y axis

we replace the unknowns in the formula for point A :

$$3 = a \cdot 2 + b$$

we do the same for point B

$$5 = a \cdot 8 + b$$

then simplify for the a unknown:

$$3 - 5 = a \cdot 2 - a \cdot 8$$

$$-2 = -6a$$

$$1/3 = a$$

then go back to one of our equations and replace a with 1/3

$$3 = 2a + b$$

$$3 = 2 \cdot (1/3) + b$$

$$3 = 2/3 + b$$

$$7/3 = b$$

and finally we get our polynom:

$$f(x) = 1/3x + 7/3$$

$$A = (2, 3)$$

$$B = (8, 5)$$

$$f(x) = \frac{1}{3}x + \frac{7}{3}$$

this is the same as `FitPoly(A,B)`



if we have 3 points, we will get a parabola as a result

$$f(x) = a \cdot x^2 + b \cdot x + c$$

$$Ay = a \cdot Ax^2 + b \cdot Ax + c$$

For n number of points, we get a polynom with (n-1) order

$$Ax = a \cdot Ax^{n-1} + b \cdot Ax^{n-2} + \dots + c \cdot Ax + d$$

Turningpoint / Extremum of a function:

$$\text{Extremum}(f)$$

Root (where $y=0$):

$$\text{Root}(f)$$

Tangent line (a line that touches the curve at a point):

first we find a point on the curve:

$$f(x) = 2x^3 - 2.5x^2 - 5x + 3$$

$$x_0 = 0.3$$



$$P = (x_0, f(x_0))$$

$$= (0.3, 1.33)$$

then we find the slope of the curve at that point, which we can get with the first derivate of f

$$\text{slope} = f'(x_0)$$

after this we have two options:

$$1. \ y - y(P) = \text{slope}(x - x(P))$$

or

$$2. \ Q = (x(P) + 1, y(P) + \text{slope})$$

$$\text{Line}(P, Q)$$

